

select: σ $\wedge$ (and), $\vee$ (or), $\neg$ (not) project: Π Cartesian product: $\times$ Join: $\bowtie$	TABLES: Student (sid:integer, sname:string, major:string) Class (cid:integer, cname: string, cdesc: string) Enrolled (sid:integer, cid: integer, esemester: string, grade: string) Building (bid: integer, bname: string) Classrooms (crid:integer, bid: integer, crfloor: int) ClassAssigned (cid: integer, crid: integer, casemester: string)
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1.Find all the student's names enrolled in CS430dl.

$\Pi_{\text{sname}}(\sigma_{\text{Student.sid}=\text{Enrolled.sid} \wedge \text{Enrolled.cid}=\text{Class.cid} \wedge \text{Class.cname}=\text{'CS430dl'}}(\text{Student} \times \text{Enrolled} \times \text{Class}))$

2.Find all the classes Hans Solo took in the SP16 semester.

$\Pi_{\text{cname}}(\sigma_{\text{Student.sid}=\text{Enrolled.sid} \wedge \text{Enrolled.cid}=\text{Class.cid} \wedge \text{Student.sname}=\text{'Hans Solo'} \wedge \text{esemester}=\text{'SP16'}}(\text{Student} \times \text{Enrolled} \times \text{Class}))$

3.Find all the classrooms on the second floor of building "A".

$\Pi_{\text{crid}}(\sigma_{\text{Classrooms.bid}=\text{Building.bid} \wedge \text{bname}=\text{'A'} \wedge \text{Classrooms.crfloor}=2}(\text{Classrooms} \times \text{Building}))$

4.Find all the class names that are located in Classroom 130.

$\Pi_{\text{cname}}(\sigma_{\text{crid}=130}(\text{Class} \bowtie_{\text{Class.cid}=\text{ClassAssigned.cid}} \text{ClassAssigned}))$

5.Find all the buildings that have ever had CS430dl in one of their classrooms.

$\Pi_{\text{bname}}(\sigma_{\text{Class.cid}=\text{ClassAssigned.cid} \wedge \text{ClassAssigned.crid}=\text{Classrooms.crid} \wedge \text{Classrooms.bid}=\text{Building.bid} \wedge \text{Class.cname}=\text{'CS430dl'}}(\text{Class} \times \text{ClassAssigned} \times \text{Classrooms} \times \text{Building}))$

6. Find all the classrooms that Alice Wonderland has been in.

$\Pi_{\text{crid}}(\sigma_{\text{sname}='Alice\ Wonderland'}($
 $(\text{Student} \bowtie_{\text{Student.sid}=\text{Enrolled.sid}} \text{Enrolled})$
 $\bowtie_{\text{Enrolled.cid}=\text{ClassAssigned.cid} \wedge \text{Enrolled.esemester}=\text{ClassAssigned.casemester}} \text{ClassAssigned})$
 $)$

7. Find all the students with a CS major that have been in a class in either the "A" building or the "B" building.

$\Pi_{\text{sname}}($
 $\sigma_{\text{Student.sid}=\text{Enrolled.sid}}$
 $\wedge \text{Enrolled.cid}=\text{ClassAssigned.cid}$
 $\wedge \text{Enrolled.esemester}=\text{ClassAssigned.casemester}$
 $\wedge \text{ClassAssigned.crid}=\text{Classrooms.crid}$
 $\wedge \text{Classrooms.bid}=\text{Building.bid}$
 $\wedge \text{Student.major}='CS'$
 $\wedge (\text{Building.bname}='A' \vee \text{Building.bname}='B') (\text{Student} \times \text{Enrolled} \times \text{ClassAssigned} \times \text{Classrooms} \times \text{Building})$
 $)$

8. Find all the classrooms that are in use during the SS21 semester.

$\Pi_{\text{crid}}(\sigma_{\text{casemester}='SS21'}(\text{ClassAssigned}))$